

Question	Answer	Marks
3(a)	(during melting,) bonds between atoms/molecules are broken	B1
	potential energy of atoms/molecules is increased	B1
	no/little work done so required input of energy is thermal	B1
3(b)(i)	$(\Delta Q =) mc\Delta\theta$	C1
	$\text{loss} = (160 \times 0.910 \times 15) + (330 \times 4.18 \times 15)$ $= 2.3 \times 10^4 \text{ J}$	A1
3(b)(ii)	$2.3 \times 10^4 = (48 \times 2.10 \times 18) + 48L + (48 \times 4.18 \times 23)$	C1
	$48L = 1.66 \times 10^4$ $L = 350 \text{ J g}^{-1}$	A1

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1(c)	acceleration proportional to displacement	D1